

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(-\frac{1}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_d^{\text{P}}] \delta_{i1i2} + \frac{5}{72} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_d^{\text{P}}] \delta_{i1i2} - \frac{4}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_d^{\text{P}}] \delta_{i1i2} - \right. \\
& \frac{1}{9} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_e^{\text{P}}] \delta_{i1i2} + \frac{5}{24} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_e^{\text{P}}] \delta_{i1i2} - \frac{4}{45} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_e^{\text{P}}] \delta_{i1i2} - \\
& \frac{1}{18} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_l^{\text{P}}] \delta_{i1i2} + \frac{5}{48} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_l^{\text{P}}] \delta_{i1i2} - \frac{2}{45} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_l^{\text{P}}] \delta_{i1i2} - \\
& \frac{1}{54} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_q^{\text{P}}] \delta_{i1i2} + \frac{5}{144} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_q^{\text{P}}] \delta_{i1i2} - \frac{2}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_q^{\text{P}}] \delta_{i1i2} - \\
& \frac{4}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0} [m_u^{\text{P}}] \delta_{i1i2} + \frac{5}{18} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1} [m_u^{\text{P}}] \delta_{i1i2} - \frac{16}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2} [m_u^{\text{P}}] \delta_{i1i2} - \\
& \frac{1}{18} g_1^4 \text{LF}_{3,0} [\tilde{\mu}] \delta_{i1i2} - \frac{1}{12} g_1^4 \text{LF}_{4,-1} [\tilde{\mu}] \delta_{i1i2} + \frac{4}{45} g_1^4 \text{LF}_{5,-2} [\tilde{\mu}] \delta_{i1i2} + \\
& \frac{1}{2} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{2,1,0} [m_1, m_e^{\text{P}}] - g_1^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{3,1,-1} [m_1, m_e^{\text{P}}] + \\
& \frac{1}{2} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{4,1,-2} [m_1, m_e^{\text{P}}] - \frac{1}{96} g_1^4 \text{LF}_{2,1,0} [m_1, m_l^{i1}] \delta_{i1i2} - \\
& \frac{1}{96} g_1^4 \text{LF}_{2,2,-1} [m_1, m_l^{i1}] \delta_{i1i2} + \frac{1}{48} g_1^4 \text{LF}_{3,1,-1} [m_1, m_l^{i1}] \delta_{i1i2} - \frac{1}{96} g_1^4 \text{LF}_{4,1,-2} [m_1, m_l^{i1}] \delta_{i1i2} - \\
& \frac{1}{96} g_1^4 \text{LF}_{2,1,0} [m_1, m_l^{i2}] \delta_{i1i2} - \frac{1}{96} g_1^4 \text{LF}_{2,2,-1} [m_1, m_l^{i2}] \delta_{i1i2} + \frac{1}{48} g_1^4 \text{LF}_{3,1,-1} [m_1, m_l^{i2}] \delta_{i1i2} - \\
& \frac{1}{96} g_1^4 \text{LF}_{4,1,-2} [m_1, m_l^{i2}] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_2, m_l^{i1}] \delta_{i1i2} - \\
& \frac{1}{32} g_1^2 g_2^2 \text{LF}_{2,2,-1} [m_2, m_l^{i1}] \delta_{i1i2} + \frac{1}{16} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_2, m_l^{i1}] \delta_{i1i2} - \\
& \frac{1}{32} g_1^2 g_2^2 \text{LF}_{4,1,-2} [m_2, m_l^{i1}] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_2, m_l^{i2}] \delta_{i1i2} - \\
& \frac{1}{32} g_1^2 g_2^2 \text{LF}_{2,2,-1} [m_2, m_l^{i2}] \delta_{i1i2} + \frac{1}{16} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_2, m_l^{i2}] \delta_{i1i2} - \\
& \frac{1}{32} g_1^2 g_2^2 \text{LF}_{4,1,-2} [m_2, m_l^{i2}] \delta_{i1i2} - \frac{1}{6} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{2,2,0} [m_d^{\text{r}}, m_q^{\text{P}}] \delta_{i1i2} + \\
& \frac{1}{12} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{3,2,-1} [m_d^{\text{r}}, m_q^{\text{P}}] \delta_{i1i2} + \frac{1}{6} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{2,1,0} [m_e^{\text{P}}, \tilde{\mu}] - \\
& \frac{1}{12} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{2,2,-1} [m_e^{\text{P}}, \tilde{\mu}] - \frac{1}{12} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{3,1,-1} [m_e^{\text{P}}, \tilde{\mu}] - \\
& \frac{1}{6} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{2,2,0} [m_e^{\text{r}}, m_l^{\text{P}}] \delta_{i1i2} + \frac{1}{12} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{3,2,-1} [m_e^{\text{r}}, m_l^{\text{P}}] \delta_{i1i2} + \\
& \frac{1}{6} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{3,1,0} [m_l^{\text{P}}, m_e^{\text{r}}] \delta_{i1i2} + \frac{1}{6} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{3,2,-1} [m_l^{\text{P}}, m_e^{\text{r}}] \delta_{i1i2} - \\
& \frac{3}{8} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{4,1,-1} [m_l^{\text{P}}, m_e^{\text{r}}] \delta_{i1i2} + \frac{1}{6} g_1^2 \overline{a}_e^{\text{pr}} a_e^{\text{pr}} \text{LF}_{5,1,-2} [m_l^{\text{P}}, m_e^{\text{r}}] \delta_{i1i2} + \\
& \frac{1}{48} g_1^4 \text{LF}_{2,1,0} [m_l^{i1}, m_1] \delta_{i1i2} - \frac{1}{96} g_1^4 \text{LF}_{3,1,-1} [m_l^{i1}, m_1] \delta_{i1i2} + \\
& \frac{1}{16} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_l^{i1}, m_2] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_l^{i1}, m_2] \delta_{i1i2} + \\
& \frac{1}{48} g_1^4 \text{LF}_{2,1,0} [m_l^{i2}, m_1] \delta_{i1i2} - \frac{1}{96} g_1^4 \text{LF}_{3,1,-1} [m_l^{i2}, m_1] \delta_{i1i2} + \\
& \frac{1}{16} g_1^2 g_2^2 \text{LF}_{2,1,0} [m_l^{i2}, m_2] \delta_{i1i2} - \frac{1}{32} g_1^2 g_2^2 \text{LF}_{3,1,-1} [m_l^{i2}, m_2] \delta_{i1i2} + \\
& \frac{1}{6} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{3,1,0} [m_q^{\text{P}}, m_d^{\text{r}}] \delta_{i1i2} + \frac{1}{6} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\text{P}}, m_d^{\text{r}}] \delta_{i1i2} - \\
& \frac{5}{8} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{4,1,-1} [m_q^{\text{P}}, m_d^{\text{r}}] \delta_{i1i2} + \frac{1}{2} g_1^2 \overline{a}_d^{\text{pr}} a_d^{\text{pr}} \text{LF}_{5,1,-2} [m_q^{\text{P}}, m_d^{\text{r}}] \delta_{i1i2} + \\
& \frac{1}{12} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{2,2,0} [m_q^{\text{P}}, m_u^{\text{r}}] \delta_{i1i2} - \frac{1}{24} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{3,2,-1} [m_q^{\text{P}}, m_u^{\text{r}}] \delta_{i1i2} - \\
& \frac{1}{12} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{3,1,0} [m_u^{\text{r}}, m_q^{\text{P}}] \delta_{i1i2} - \frac{1}{12} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{3,2,-1} [m_u^{\text{r}}, m_q^{\text{P}}] \delta_{i1i2} - \\
& \frac{1}{4} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{4,1,-1} [m_u^{\text{r}}, m_q^{\text{P}}] \delta_{i1i2} + \frac{1}{2} g_1^2 \tilde{\mu}^2 \overline{y}_u^{\text{pr}} y_u^{\text{pr}} \text{LF}_{5,1,-2} [m_u^{\text{r}}, m_q^{\text{P}}] \delta_{i1i2} + \\
& \frac{5}{24} g_1^4 \text{LF}_{4,1,-2} [\tilde{\mu}, m_1] \delta_{i1i2} - \frac{1}{6} g_1^4 \text{LF}_{5,1,-3} [\tilde{\mu}, m_1] \delta_{i1i2} + \frac{5}{8} g_1^2 g_2^2 \text{LF}_{4,1,-2} [\tilde{\mu}, m_2] \delta_{i1i2} - \\
& \frac{1}{2} g_1^2 g_2^2 \text{LF}_{5,1,-3} [\tilde{\mu}, m_2] \delta_{i1i2} - \frac{1}{12} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{2,1,0} [\tilde{\mu}, m_e^{\text{P}}] + \\
& \frac{1}{24} g_1^2 \overline{y}_e^{i2p} y_e^{i1p} \text{LF}_{3,1,-1} [\tilde{\$$